INCH-POUND

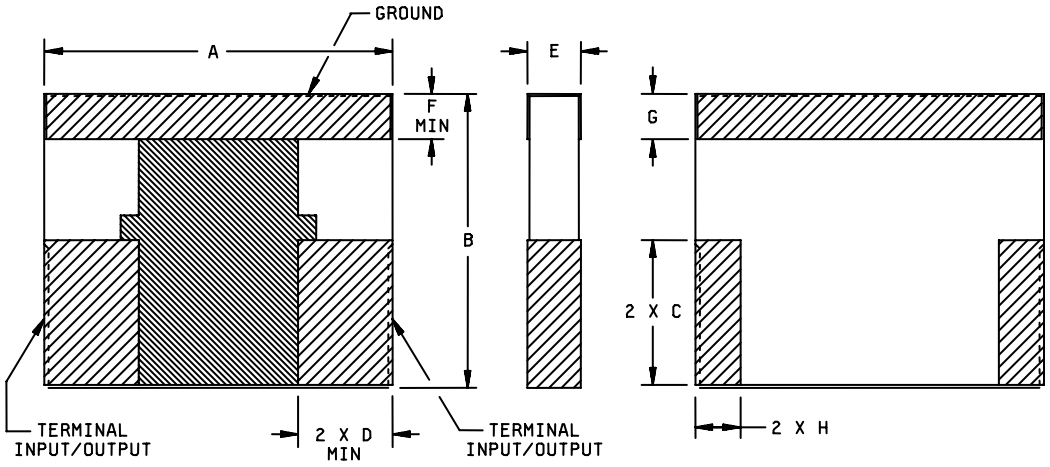
MIL-DTL-3933/29
1 August 2006

DETAIL SPECIFICATION SHEET

ATTENUATORS, FIXED, CHIP, (SURFACE MOUNT),
0-20 dB, FREQUENCY RANGE: DC TO 18 GHZ

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-3933.




| Symbol | Inches | | mm | |
|---|---|---|---|---|
| | Min | Max | Min | Max |
| A | .145 | .149 | 3.68 | 3.78 |
| B | .120 | .124 | 3.05 | 3.15 |
| C | .059 | .063 | 1.50 | 1.60 |
| D | .0285 | .0385 | 0.724 | 0.978 |
| E | .019 | .023 | 0.48 | 0.58 |
| F | .013 | .017 | 0.33 | 0.43 |
| G | .013 | .017 | 0.33 | 0.43 |
| H | .013 | .017 | 0.33 | 0.43 |

NOTES:

1. Dimensions are in inches; metric equivalents are in parentheses.
2. Laser mark the dB value using height of .010.
3. Tolerances: Decimals: ± .010 (0.25 mm) for one decimal point, ± .005 (0.13 mm) for two decimal points and ± .002 (0.05 mm) for three decimal points.
4. Topology of resistor is determined by independent manufacturer's design.

FIGURE 1. Dimensions and configurations for PIN M3933/29-01 through M3933/29-40.



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ENGINEERING DATA:

Drop-in surface mountable chip attenuators for use in general field equipment.

Part or Identifying Number (PIN): PINs to be used for surface mount chip fixed attenuators acquired to this specification sheet are created as follows:

| M | 3933 | /29- | 01 | X | $Y_1Y_2$ | Z |
|---|---|---|---|---|---|---|
| M prefix | Specification number | Specification sheet number | Dash number | Suffix letter | Critical interface identifier | Quality level class K or H |

Suffix letter (as assigned in QPL-3933): The suffix letter in the above PIN shall be incorporated to denote the chip manufacturer.

Critical interface identifier: Assigned consistent with MIL-DTL-3933. $Y_1$ should identify what metal is used: aluminum (A), copper (C), gold (G), nickel (N); $Y_2$ should identify body style used: open bond pads (P), welded leads already attached (L), wrap around leads (W).

Quality level: Class K or H is assigned consistent with MIL-DTL-3933 Appendix C.

Materials:

Substrate: 99.5 percent alumina.

Resistive film: Tantalum nitride stabilized for 30 minutes minimum at 300°C minimum.

Terminal pads: Copper, nickel and gold.

Finishes:

Terminal: Final coating is Sn62 and maybe fused or solder dipped (as restricted, see 1/ below).

Marking:

Substrate: Laser mark attenuator slash value as shown in figure 1 on top and bottom of substrate. Height of marking shall be .010 inch (0.25 mm). Also, laser mark “suffix letter” (from PIN above).

Operating frequency range: DC to 18 gigahertz (GHz).

Operating temperature range: -55°C to +125°C. Note: Power input is derated linearly to 0.5 watt at +125°C.

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1/ The use of pure tin, as an underplate or final finish, is prohibited both internally and externally. Tin content of surface mount chip fixed attenuator components and solder shall not exceed 97 percent, by mass. Tin shall be alloyed with a minimum of 3 percent lead, by mass. The use of alloys with tin content greater than 97 percent, by mass, may exhibit tin whisker growth problems after manufacture. Tin whiskers may occur anytime from a day to years after manufacture and can develop under typical operating conditions, on products that use such materials. Conformal coatings applied over top of a whisker-prone surface will not prevent the formation of tin whiskers. Alloys of 3 percent lead, by mass, have been shown to inhibit the growth of tin whiskers. For additional information on this matter, refer to ASTM-B545 (Standard Specification for Electrodeposited Coatings of Tin).



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REQUIREMENTS:

Dimensions and configuration: See figure 1.

Nominal impedance: 50 ohms.

Attenuation: See table I. Stability and sensitivity:

| **Maximum change in attenuation after:** | **Attenuation change maximum (dB)** |
|---|---|
| Temperature change or thermal shock | 0.5 |
| Vibration or shock | 0.5 |
| Moisture resistance or salt spray | 0.5 |
| Peak power | 0.5 |

Power: Maximum power output.

AV (W) at 25°C (continuously): 2.0 Watts.

Peak (W): 50 Watts (Duty cycle 5 x $10^{-4}$; maximum pulse duration of 5 μsec).

Power sensitivity of attenuation, maximum: 0.005 dB/dB/Watt (measured at DC).

Temperature sensitivity of attenuation, maximum: 0.0006 dB/dB/°C (measured at DC).

TABLE I. Electrical characteristics.

| Dash number | Attenuation (dB) | | | | | Maximum power input | | VSWR maximum | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | Nominal | Deviation (±) | | | | AV (W) at 25°C 1/ (contin-uously) | Peak 2/ (W) | DC to 4 GHz | 4 to 8 GHz | 8 to 12.4 GHz | 12.4 to 18 GHz |
| | | DC to 4 GHz | 4 to 8 GHz | 8 to 12.4 GHz | 12.4 to 18 GHz | | | | | | |
| 01 K H | 0.5 | 0.3 | 0.4 | 0.6 | 1.1 | 2.0 | 50 | 1.2:1 | 1.25:1 | 1.35:1 | 1.55:1 |
| 02 K H | 1.0 | 0.3 | 0.4 | 0.6 | 1.1 | 2.0 | 50 | 1.2:1 | 1.25:1 | 1.35:1 | 1.55:1 |
| 03 K H | 1.5 | 0.3 | 0.4 | 0.6 | 1.1 | 2.0 | 50 | 1.2:1 | 1.25:1 | 1.35:1 | 1.55:1 |
| 04 K H | 2.0 | 0.3 | 0.4 | 0.6 | 1.1 | 2.0 | 50 | 1.2:1 | 1.25:1 | 1.35:1 | 1.55:1 |
| 05 K H | 2.5 | 0.3 | 0.4 | 0.6 | 1.1 | 2.0 | 50 | 1.2:1 | 1.25:1 | 1.35:1 | 1.55:1 |
| 06 K H | 3.0 | 0.3 | 0.4 | 0.6 | 1.1 | 2.0 | 50 | 1.2:1 | 1.25:1 | 1.35:1 | 1.55:1 |
| 07 K H | 3.5 | 0.3 | 0.4 | 0.6 | 1.1 | 2.0 | 50 | 1.2:1 | 1.25:1 | 1.35:1 | 1.55:1 |
| 08 K H | 4.0 | 0.3 | 0.4 | 0.6 | 1.1 | 2.0 | 50 | 1.2:1 | 1.25:1 | 1.35:1 | 1.55:1 |
| 09 K H | 4.5 | 0.3 | 0.4 | 0.6 | 1.1 | 2.0 | 50 | 1.2:1 | 1.25:1 | 1.35:1 | 1.55:1 |
| 10 K H | 5.0 | 0.3 | 0.4 | 0.6 | 1.1 | 2.0 | 50 | 1.2:1 | 1.25:1 | 1.35:1 | 1.55:1 |
| 11 K H | 5.5 | 0.3 | 0.4 | 0.6 | 1.1 | 2.0 | 50 | 1.2:1 | 1.25:1 | 1.35:1 | 1.55:1 |
| 12 K H | 6.0 | 0.3 | 0.4 | 0.6 | 1.1 | 2.0 | 50 | 1.2:1 | 1.25:1 | 1.35:1 | 1.55:1 |
| 13 K H | 6.5 | 0.3 | 0.4 | 0.6 | 1.1 | 2.0 | 50 | 1.2:1 | 1.25:1 | 1.35:1 | 1.55:1 |



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TABLE I. Electrical characteristics - Continued.

| Dash number | Attenuation (dB) | | | | | Maximum power input | | VSWR maximum | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | Nominal | Deviation (±) | | | | AV (W) at 25°C 1/ (contin-uously) | Peak 2/ (W) | DC to 4 GHz | 4 to 8 GHz | 8 to 12.4 GHz | 12.4 to 18 GHz |
| | | DC to 4 GHz | 4 to 8 GHz | 8 to 12.4 GHz | 12.4 to 18 GHz | | | | | | |
| 14 KH | 7.0 | 0.3 | 0.4 | 0.6 | 1.1 | 2.0 | 50 | 1.2:1 | 1.25:1 | 1.35:1 | 1.55:1 |
| 15 KH | 7.5 | 0.3 | 0.4 | 0.6 | 1.1 | 2.0 | 50 | 1.2:1 | 1.25:1 | 1.35:1 | 1.55:1 |
| 16 KH | 8.0 | 0.3 | 0.4 | 0.6 | 1.1 | 2.0 | 50 | 1.2:1 | 1.25:1 | 1.4:1 | 1.65:1 |
| 17 KH | 8.5 | 0.3 | 0.4 | 0.6 | 1.1 | 2.0 | 50 | 1.2:1 | 1.25:1 | 1.4:1 | 1.65:1 |
| 18 KH | 9.0 | 0.3 | 0.4 | 0.6 | 1.1 | 2.0 | 50 | 1.2:1 | 1.25:1 | 1.4:1 | 1.65:1 |
| 19 KH | 9.5 | 0.3 | 0.4 | 0.6 | 1.1 | 2.0 | 50 | 1.2:1 | 1.25:1 | 1.4:1 | 1.65:1 |
| 20 KH | 10.0 | 0.3 | 0.4 | 0.6 | 1.1 | 2.0 | 50 | 1.2:1 | 1.25:1 | 1.4:1 | 1.65:1 |
| 21 KH | 10.5 | 0.3 | 0.4 | 0.6 | 1.1 | 2.0 | 50 | 1.2:1 | 1.25:1 | 1.4:1 | 1.65:1 |
| 22 KH | 11.0 | 0.3 | 0.4 | 0.6 | 1.1 | 2.0 | 50 | 1.2:1 | 1.25:1 | 1.4:1 | 1.65:1 |
| 23 KH | 11.5 | 0.3 | 0.4 | 0.6 | 1.1 | 2.0 | 50 | 1.2:1 | 1.25:1 | 1.4:1 | 1.65:1 |
| 24 KH | 12.0 | 0.3 | 0.4 | 0.6 | 1.1 | 2.0 | 50 | 1.2:1 | 1.25:1 | 1.4:1 | 1.65:1 |
| 25 KH | 12.5 | 0.3 | 0.4 | 0.6 | 1.1 | 2.0 | 50 | 1.2:1 | 1.25:1 | 1.4:1 | 1.65:1 |
| 26 KH | 13.0 | 0.3 | 0.4 | 0.6 | 1.1 | 2.0 | 50 | 1.2:1 | 1.25:1 | 1.4:1 | 1.65:1 |
| 27 KH | 13.5 | 0.3 | 0.4 | 0.6 | 1.1 | 2.0 | 50 | 1.2:1 | 1.25:1 | 1.4:1 | 1.65:1 |
| 28 KH | 14.0 | 0.3 | 0.4 | 0.6 | 1.1 | 2.0 | 50 | 1.2:1 | 1.25:1 | 1.4:1 | 1.65:1 |
| 29 KH | 14.5 | 0.3 | 0.4 | 0.6 | 1.1 | 2.0 | 50 | 1.2:1 | 1.25:1 | 1.4:1 | 1.65:1 |
| 30 KH | 15.0 | 0.3 | 0.4 | 0.6 | 1.1 | 2.0 | 50 | 1.2:1 | 1.25:1 | 1.4:1 | 1.65:1 |
| 31 KH | 15.5 | 0.3 | 0.4 | 0.6 | 1.1 | 2.0 | 50 | 1.2:1 | 1.25:1 | 1.4:1 | 1.65:1 |
| 32 KH | 16.0 | 1.0 | 1.0 | 2.5 | 5 | 2.0 | 50 | 1.2:1 | 1.25:1 | 1.4:1 | 1.65:1 |
| 33 KH | 16.5 | 1.0 | 1.0 | 2.5 | 5 | 2.0 | 50 | 1.2:1 | 1.25:1 | 1.4:1 | 1.65:1 |
| 34 KH | 17.0 | 1.0 | 1.0 | 2.5 | 5 | 2.0 | 50 | 1.2:1 | 1.25:1 | 1.4:1 | 1.65:1 |
| 35 KH | 17.5 | 1.0 | 1.0 | 2.5 | 5 | 2.0 | 50 | 1.2:1 | 1.25:1 | 1.4:1 | 1.65:1 |
| 36 KH | 18.0 | 1.0 | 1.0 | 2.5 | 5 | 2.0 | 50 | 1.2:1 | 1.25:1 | 1.4:1 | 1.65:1 |
| 37 KH | 18.5 | 1.0 | 1.0 | 2.5 | 5 | 2.0 | 50 | 1.2:1 | 1.25:1 | 1.4:1 | 1.65:1 |
| 38 KH | 19.0 | 1.0 | 1.0 | 2.5 | 5 | 2.0 | 50 | 1.2:1 | 1.25:1 | 1.4:1 | 1.65:1 |
| 39 KH | 19.5 | 1.0 | 1.0 | 2.5 | 5 | 2.0 | 50 | 1.2:1 | 1.25:1 | 1.4:1 | 1.65:1 |
| 40 KH | 20.0 | 1.0 | 1.0 | 2.5 | 5 | 2.0 | 50 | 1.2:1 | 1.25:1 | 1.4:1 | 1.65:1 |

1/ Average power is linearly derated to 0.5 watt at +125°C.

2/ Duty cycle 5 x $10^{-4}$; maximum pulse duration of 5 μsec.



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Referenced documents. In addition to MIL-DTL-3933, this document references the following:

ASTM-B545

Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Preparing activity:
DLA - CC

(Project 5985-2006-008)

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